

Graph Neural Networks for Drug–Drug Interaction Prediction—Predicting Safe Drug Pairings with AI [†]

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Abstract

At present, polypharmacy—which is the use of several medications to treat a single case at the same time—has become a fairly common medical practice, particularly in chronic illnesses or with older patients. But this relatively ‘faster’ form of treatment brings the problem of cumulative polypharmacy, which occurs when there is an increase in drug–drug interactions (DDIs) due to the large number of medicines taken. While the aftermath, such as the reduction in strength of medication taken or catastrophic and fatal responses to certain drugs, is clearly not worth the initial effort put into trying to ease the condition, attempting to resolve these issues requires excessive research. With these difficulties in mind, we describe our research that uses graph neural networks (GNNs) focused on DDI prediction by modeling drugs and their interactions in the form of graphs. The research is divided into two parts. In this research, the relevant literature is reviewed in order to understand how modern GNN-based algorithms can be applied for the detection of optimal drugs.

Keywords: graph neural networks; GraphRX; catastrophic; cumulative polypharmacy



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1. Introduction

The growing practice of polypharmacy, especially among the elderly and patients with chronic conditions, has come with rising worries concerning drug–drug interactions (DDIs) [1]. Such interactions may modify the effects of drugs as intended, bringing about undesirable adverse reactions or even poor drug efficacy. According to the World Health Organization, DDIs account for 20% of ADRs (adverse drug reactions), the majority of which are avoidable. For this reason, there is an urgent need for robust, scalable models for DDI prediction in order to make it possible for practitioners to take measures against harmful drug interactions [2]. In light of the problem, graph neural networks (GNNs) may provide a better approach. GNNs treat drug interactions not as individual phenomena; rather, GNNs use the interactions among drugs, which are seen as nodes in a network with edges depicting interactions. With such an approach, GNNs manage to model the direct and indirect relationships between drugs, resulting in more accurate and interpretable DDI predictions [3]. This research, named GraphRX, focuses on creating a model that can predict DDIs based on graph neural networks (algorithm that combines multiple layers of

data synthesis). The main aim is to help physicians determine which combination of drugs should most likely be prescribed for the benefit of the patient. This discusses the research context, research problem, and research methods to be used during the process.

These past few years have seen the emergence of predictive models based on machine learning approaches to predicting DDIs. [4,5] Methods such as decision trees, support vector machines, and random forests have seen some success in predicting interactions based on querying drug information datasets of larger scales. Still, these models have certain limitations, especially the aspect of being able to generalize across drug classes and the issue of the complexity of the available dimensional data.

Despite these challenges, graph neural networks systematically surpass the state of the art. GNNs, unlike the typical approach in the field, where drugs are seen as separate entities, consider them to be nodes in a broader graph. As a result, they are capable of identifying both direct and indirect interactions between drugs, which makes them ideal candidates for predicting drug–drug interactions. Because of this graph representation, GNNs have the capacity to integrate even more features like chemical structure, molecular characteristics, and known relationships in their inference process. This study extends the work that already exists in the literature by focusing on an open access interaction database and implementing GNNs. Within the scope of this research, it is expected that machine learning frameworks will be able to increase the precision of DDI predictions and patients can be protected from dangerous drug combinations [6–8].

Multiple medication usage is commonly associated with drug interactions that pose a risk to the patient. Drug interactions are difficult to predict due to numerous drug combinations and the interplay of different factors that determine the interaction. CDDS(Clinical Decision Support System), guidelines, or pharmacokinetics databases are currently the only methods available for predicting DDIs, but they remain unable to do so where the interaction is novel or involves an unduly studied drug since they are retrospective. In addition, baseline machine learning algorithms find it difficult to explain in a perfect manner the relationships that exist between drug transfers, since those relations are complex and non-linear in nature. Each interaction is commonly seen as an isolated event without the consideration of possible interactions of a drug combination which results in poor generalization, yielding false negatives and positives. These challenges are particularly pronounced for drug pair combinations with limited exposure in the scientific literature. This research seeks to address the following research challenge:

1. How can we leverage graph neural networks (GNNs) to more accurately predict drug–drug interactions (DDIs)?
2. What are the key factors that contribute to the safe pairing of drugs, and how can these be incorporated into a predictive model?

By responding to these questions, the research seeks to identify a better and more predictive DDI method so as to assist patients, thereby reducing the chances of suffering from a combination of undesirable drug effects.

The contributions of the research have both theoretic and practical relevance in the fields of academia and healthcare. By the end of the study, the following main outputs will be available: The GNN model for DDI prediction will be fully trained and applied. This will be an enhancement of the other previously used models in the sense that the model will use a new method of predicting interactions of drugs that will harm patients. Web-based DDI prediction platform: This will be useful for healthcare practitioners, as it will allow them to determine the safety of a certain combination of drugs in real time. Assessment and evaluation of results: In one or two months, this task will involve the evaluation of the GNN model with a focus on its accuracy/precision and performance in interrelations of drugs. Outcome of the study and recommendations for further studies: It is expected that

the study will expand the existing knowledge in DDI prediction, and it will be published in the form of a research paper that relates to DDI.

2. Literature Review

Almost any modern pharmacological agent has the potential of interaction with other agents, and hence drug–drug interactions (DDIs) become a significant problem in pharmacology, with the most negative effects making up a large percentage of overall healthcare costs and a large volume of hospital admissions globally [9]. For many chronically ill patients and for elderly individuals to whom this issue is even more relevant, polypharmacy is the norm, and consequently the problem of DDIs becomes more acute. These interactions can lead to loss of therapeutic benefit or undesirable side effects, both of which create a risk to patient safety [10].

Traditionally, DDI prediction has used databases with pharmacological information such as DrugBank and the results of clinical studies testing drugs in combination [11–13]. However, this approach has its shortcomings: the data is retrospective and is limited to a few combinations of drugs because few drug treatments have been used at that time point. As the arsenal of approved drugs increases, conducting clinical trials to derive possible combinations and interactions becomes increasingly impractical. So, there is a gap to fill, namely an AI-enabled system able to flag potential DDIs in advance, long before they would already be present in patients in clinics [14]. The last few years have seen some considerable advances in the field of machine learning, which has been accompanied by the development of computational models that are capable of predicting DDIs from the interactions of many drugs in a large-scale data repository. These models employ different DDI prediction techniques incorporating supervised, unsupervised, and reinforcement learning to extract patterns representative of potential interaction. However, most of these models utilize feature-based representations of drugs to construct the DDI predictions, which is a major hindrance to drug interaction predictions since these representations do not capture the drug's molecular interaction complexities [15].

Machine learning techniques began to enter the picture during the last decade, and these technologies are evolving in the DDI domain. Furthermore, quite a few models were proposed for prediction such as decision trees, random forest, or SVM(Support Vector Machine)-based algorithms [16,17]. They frame the DDI prediction task as a binary classification problem, where the task is to determine whether a contact between a given pair of drugs will take place (i.e., in the positive class) or will not take place (i.e., in the negative class). For example, a study suggested a random forest model which was guided by multiple properties of drugs like chemical structure, target proteins, and metabolic pathways in the prediction of DDIs. While it is known that this model was able to predict correctly known interactions, it failed to predict such drug combinations not represented in the training set from previously unseen interaction data [18]. Moreover, the model experienced difficulties with high-dimensional feature spaces which yielded challenges to the interpretation of the findings and to gaining understanding of the interactions' mechanics. The application of support vector machines (SVMs) for DDI detection is another widely practiced scheme. SVMs have been applied in various studies on predicting DDIs by using drug features and placing them in a high-dimensional metric space with interactions being hyperplanes separating interacting and non-interacting drug pairs. Nonetheless, similar to random forests, SVMs also do not perform well in defining the intricate non-linear modalities in drug interactions, which leads to false positive and negative error rates that are too high.

There also remains the problem of generalization. GNNs have performed well when predicting known DDIs, but they tend to find it hard to predict interactions which include

new drugs. This is especially the case when considering drug development processes, because one would want to be able to predict the interactions of new drugs that have not been tried out. In response to this problem, researchers are seeking new model architecture and training strategies that improve the extent of generalization of GNNs.

A comparison of the algorithms applied to DDI prediction is presented in Table 1.

Table 1. Comparison of GNN algorithms for DDI prediction.

Algorithm	Strengths	Limitations	Source
Graph Convolutional Networks (GCNs)	Molecular graph networks are able to obtain local neighborhood information efficiently and are therefore frequently applied to molecular prediction tasks.	Fairly limited in their scalability; they do however seem to struggle with very large datasets.	[19]
Graph Attention Networks (GATs)	Improved accuracy through attention mechanisms, better for complex relationships.	Lacking efficiency in computations because necessary weights must be incorporated for each attention neighbor.	[20]
Message-Passing Neural Networks (MPNNs)	Excellent at capturing complex chemical properties, flexible message-passing.	High computational cost; difficult to scale to large datasets.	[21]
GraphSAGE	Highly scalable, can handle large datasets efficiently, generalizes well to unseen graphs.	May lose information from distant nodes; focuses primarily on local structures.	[22]
DeepChem	Combines GNNs with deep learning techniques, designed for large pharmaceutical datasets.	Limited real-time application due to its high computational cost.	[23]

The Figure 1 illustrates the fundamental difference between a standard convolution operation used in image processing and a graph convolution operation used in GNN. On the left, the standard convolution is shown on a grid-like structure, similar to the pixels of an image. The central red node represents the pixel being updated. It gathers information from its immediately surrounding neighbors (the gray nodes) in a very structured, local, and uniform manner.

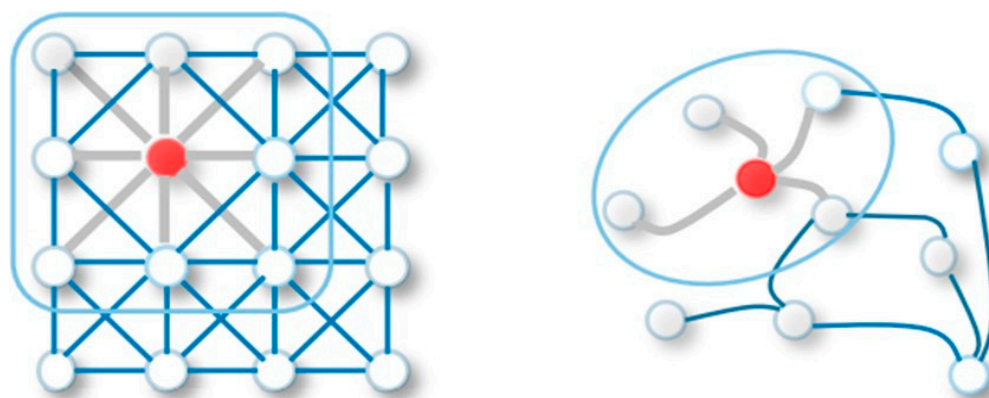


Figure 1. Convolution vs. graph convolution [23].

3. Methodology

The study in this instance adopts a quantitative, experimental design with which a GNN-based predictive model is built and validated [24]. This design is relevant to this entire project because it provides the ability to objectively evaluate the model using clear

metrics and specify the degree to which the goals are met. Experimental designs are important in this instance since they allow for the manipulation of variables followed by certain outcomes, which is an excellent method of understanding how well the GNN will be able to predict the DDIs [25].

3.1. Data Collection

The collection of data is central to this project. It is important to stress here that the model has to be trained on as complete a dataset as possible in order to achieve maximum effectiveness in the prediction. The main sources of data include

1. DrugBank: A very popular and reliable database resource that has useful information on drugs including their chemical structure, their pharmacokinetics, and other drugs that are known to interact with them. DrugBank is a primary dataset for this study, as it offers the data needed to practically simulate drug interaction modeling.
2. TWOSIDES: This resource specializes exclusively on adverse effects due to drug-polypharmacy and provides additional information on the safety profile of several drugs with TWOSIDES being provided. TWOSIDES is useful in determining possible dangers that may arise from the combination of drugs in polypharmacy and provides a predictive scope for both the likelihood and the level of adverse interactions.
3. ChEMBL: The bioactivity database that ELIXIR API has includes drug interaction information, the biological effects of drugs, and drug targets. As mentioned in prior studies, ChEMBL contributes additional data related to pharmacology in this dataset, assisting the model in the comprehension of targeted system interaction at the molecular level.

The use of these sources guarantees that the created baseline will be distributed in several domains related to drug interactions, increasing the effectiveness of the model.

3.2. Data Preprocessing

The next step after data collection is data preprocessing of the entire dataset to ensure speedy and efficient analysis and model development. In this phase, there are a number of vital activities that are executed.

1. Data cleaning involves relatively more technical elements such as the rectification of errors and ensuring as much homogeneity as possible of within the dataset. As Figure 2 illustrates, raw datasets often contain inconsistencies that require careful cleaning before analysis.

	drug1_id	drug2_id	drug1_name	drug2_name	interaction_type
0	DB00006	DB00346	Bivalirudin	Alfuzosin	serum concentration
1	DB00006	DB13783	Bivalirudin	Acemetacin	risk or severity of bleeding
2	DB00006	DB06605	Bivalirudin	Apixaban	anticoagulant activities
3	DB00006	DB06695	Bivalirudin	Dabigatran etexilate	anticoagulant activities
4	DB00006	DB09075	Bivalirudin	Edoxaban	anticoagulant activities

Figure 2. Raw dataset overview.

2. Duplicate Removal: Duplicates of the samples are identified and removed to avoid any duplication that would yield incorrect results when training the model.
3. Inconsistency Resolution: There will always be variation in the names of drugs, measurement units, and types of interactions. These forms of variations are eliminated through standardization techniques.

4. Normalization: All volumetric and activity score continuous variables will be brought to standard ranges such as 0 and 1. This is important, as failure to apply normalization in models often causes bias to the model, as features with varying scales do not carry equal importance when constructing the model, which is the case in this model as well.

4. Results and Discussion

The topic of drug–drug interaction (DDI) prediction has become a central focus in medical research, with implications that extend to patient well-being, drug efficiency, and personalized medicine development. The importance of this study is that it fills traditional methods' gaps by integrating molecular data and adopting advanced graph neural network (GNN) models to predict interactions more accurately. To provide an extensive analysis of outcomes, this section discusses processes such as dataset preparation, molecular feature extraction, model implementation, evaluation, and system integration.

The first step was to carefully curate and preprocess datasets from credible repositories like DrugBank, TWOSIDES, and ChEMBL. RDKit and PubChemPy tools were used for cleaning out these databases, as well as for normalizing them and extracting features from the molecules contained in them. As a result, four advanced GNN models—namely graph convolutional networks (GCNs), graph attention networks (GATs), GraphSAGE, and message-passing neural networks (MPNNs)—use the processed data as input. In order to arrive at optimal settings for each model, there is a systematic process of hyperparameter tuning using rigorous training and evaluation methods. Interaction types having no clinical relevance such as minor gastrointestinal effects were excluded. The dataset was further refined to focus on interactions that have well-documented pharmacological consequences, such as the inhibition of CYP (cytochrome P450) enzymes or protein-binding displacement.

4.1. Representation of Inputs and Graph Construction

One of the major steps was representing drugs and interactions as a graph. The nodes in the graph represent individual drugs, while edges show interactions between them. For connectivity, an adjacency matrix was defined, where a value of 1 indicated an interaction, while 0 meant no interaction. Other molecular descriptors like logP, hydrogen bond donors/acceptors, and molecular weight were included in each node (drug), which were calculated using RDKit (2023.09.5). This information helped to encode key drug properties into these attributes. Additional data such as interaction types were added to the edges to contextualize the modeled relationships. As shown in Figure 3, the distribution of interaction types highlights the relative frequency of different categories.

Each model was adapted to ensure it predicts DDIs more efficiently. Graph convolutional networks employed spectral convolutions that merge features from adjacent nodes to focus on local interactions. GATs have attention mechanisms, allowing them to assign weights to edges and hence make this particular model concentrate only on vital drug interactions. GraphSAGE used inductive learning, where the features were sampled and aggregated by a node's neighbors, making it a good performer for large graphs. MPNNs had message-passing layers in which nodes exchange information iteratively with their neighbor, enabling the model to capture both direct and indirect interactions. These designs were aimed at effectively modeling the different complexities of DDIs. These designs were aimed at effectively modeling the different complexities of DDIs, as illustrated in Figure 4, which provides a visualization of the drug interaction graph. Figure 5, demonstrates GCN performs better than MPNN on accuracy metrics.

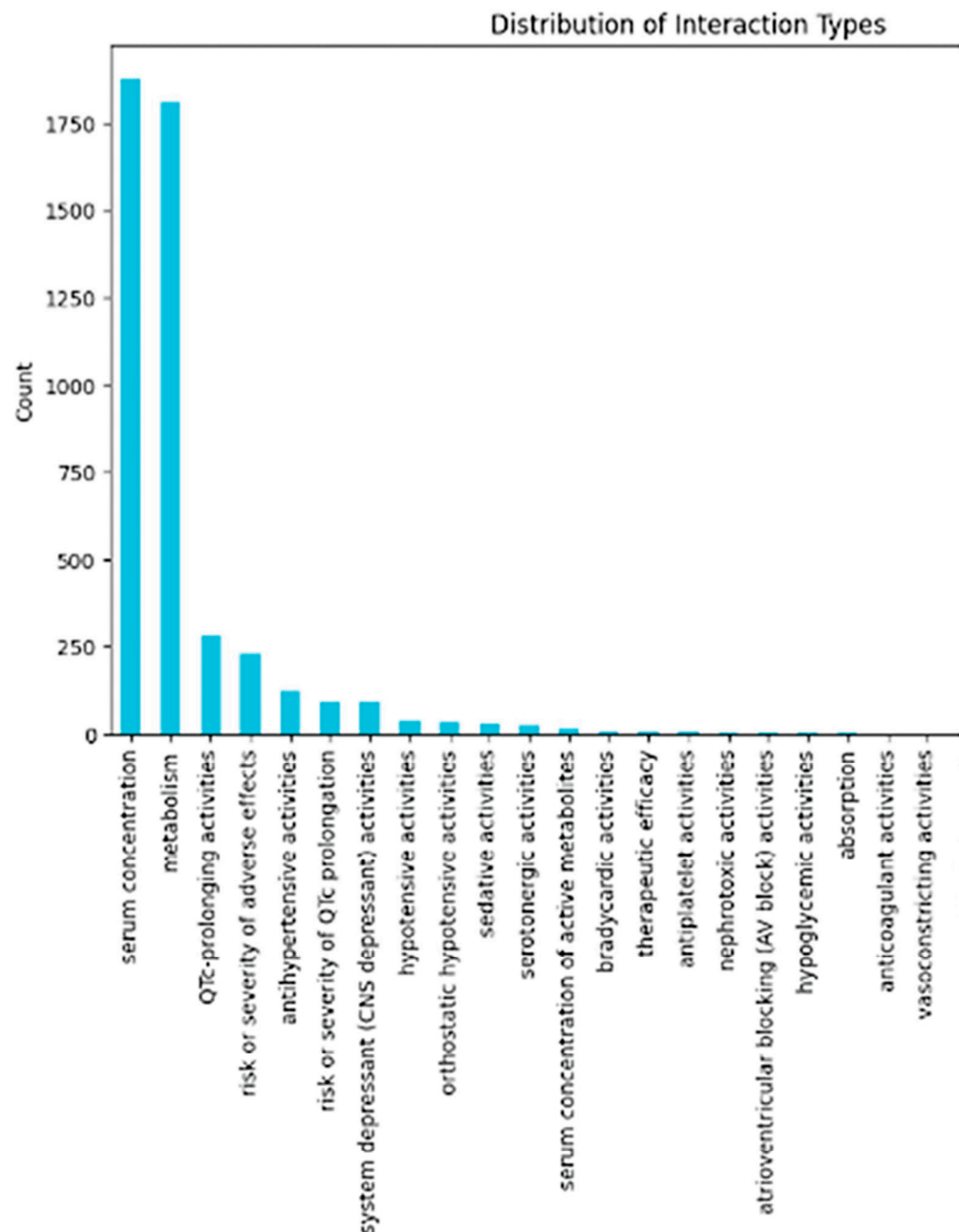


Figure 3. Distribution of interaction types.

4.2. Evaluation Metrics and Results

To measure the quality of the models, various evaluation metrics were used including AUC(Area Under The Curve), F1 score, recall, precision, and accuracy. GraphSAGE was found to be the best among these models with an accuracy of 61.04%, F1 score of 44.89%, and AUC of 56%. ROC curves for each model revealed their ability to differentiate between interacting drug pairs from non-interacting ones. GCNs and GATs performed well but lagged behind GraphSAGE in dealing with large complex graphs. MPNNs, on the other hand, were expressive but computationally expensive and prone to overfitting. Comparing these models highlighted their strengths and weaknesses. The fact that GraphSAGE is scalable and able to generalize well on unseen data made it a good choice for this experiment. On the other hand, GATs come with additional interpretability through attention mechanisms at the cost of higher computation resources as compared to others such as GCNs, which are faster but not very effective when capturing indirect interactions. These findings led to adopting GraphSAGE as the main architecture for our final model,

4.3. Frontend Design and User Interface

This acts as a bridge between complex backend operations and an intuitive visual interface that can be easily understood by people in medical settings, who are the target group, hence ensuring smooth usage. Users enter drug names into simple text boxes while receiving predictive results instantly. These include interaction predictions, confidence scores, and severity levels, which are displayed in a concise manner. The interface is completely responsive, and it adjusts depending on the device being used. This guarantees that any user can access it irrespective of their desktops, tablets, or smartphones. There are several interactive components such as a chatbot, medication reminders, and downloadable reports for enhancing user experience. These features make this app a fully fledged healthcare tool.

A major stride in this project was the ability to visualize drug structures as molecular graphs. Using RDKit, each drug was represented as a graph, where atoms served as nodes and bonds were the edges. For instance, this visualization portrayed aspirin's molecular structure (CC(=O)OC1=CC=CC=C1C(=O)O), highlighting its functional groups and possible binding sites. From such a visualization, it was evident that certain molecular features such as carboxylic groups or aromatic rings had effects with regard to drug interactions. A number of clustering algorithms such as K-means are applied in order to group drugs on the basis of their molecular features. This analysis identified clusters of drugs with similar interaction profiles, such as anticoagulants or antiplatelet agents. By visualizing these clusters, it was possible to infer common interaction mechanisms, aiding in the identification of high-risk drug combinations.

Results as shown in Figure 6, without hyperparameter tuning demonstrate the initial performance of the graph neural network (GNN) models—GCN, GAT, GraphSAGE, and MPNN—through metrics including accuracy, precision, recall, and F1 score. These were computed for each model's performance measurement. Accuracy: By achieving the highest accuracy, the GCN (65.85%) outperformed GAT (49.86%), GraphSAGE (61.04%), and MPNN (37.66%). In other words, this suggests that the GCN is able to effectively capture drug interaction patterns even without any modifications in its default configuration. Precision: All models had high precision values, with GCN having 91.25%, GAT having 92.57%, GraphSAGE having 89.57%, and MPNN having 92.12%, meaning they can minimize false positives. Recall: It was hard to find 'true positive' interactions, because recall values were significantly low. The GCN had the highest recall (18.82%), while the MPNN had the lowest (6.25%). F1 Score: The overall model performance for all models measured by the F1 score, which is a combination of precision and recall in terms of accuracy, was relatively poor, with the GCN showing the highest at 17.84%. This reflects that imbalanced datasets pose difficulties when it comes to accurate predictions of interactions. As shown in Figure 7, the drug interaction prediction model provides interactive results, including the confidence score and severity of the potential interaction. The comparison of pre-tuning and post-tuning results for GCN, GAT, GraphSAGE, and MPNN models is presented in Table 2. The inter-feature dependencies are visualized in the correlation heatmap (Figure 8).

Table 2. Comparison of pre-tuning and post-tuning accuracy and F1 scores across GNN models.

Model	Pre-Tuning Accuracy (%)	Post-Tuning Accuracy (%)	Pre-Tuning F1 (%)
GCN	65.85	39.80	17.84
GAT	49.86	43.70	12.38
GraphSAGE	61.04	44.89	17.28
MPNN	37.66	7.09	3.85

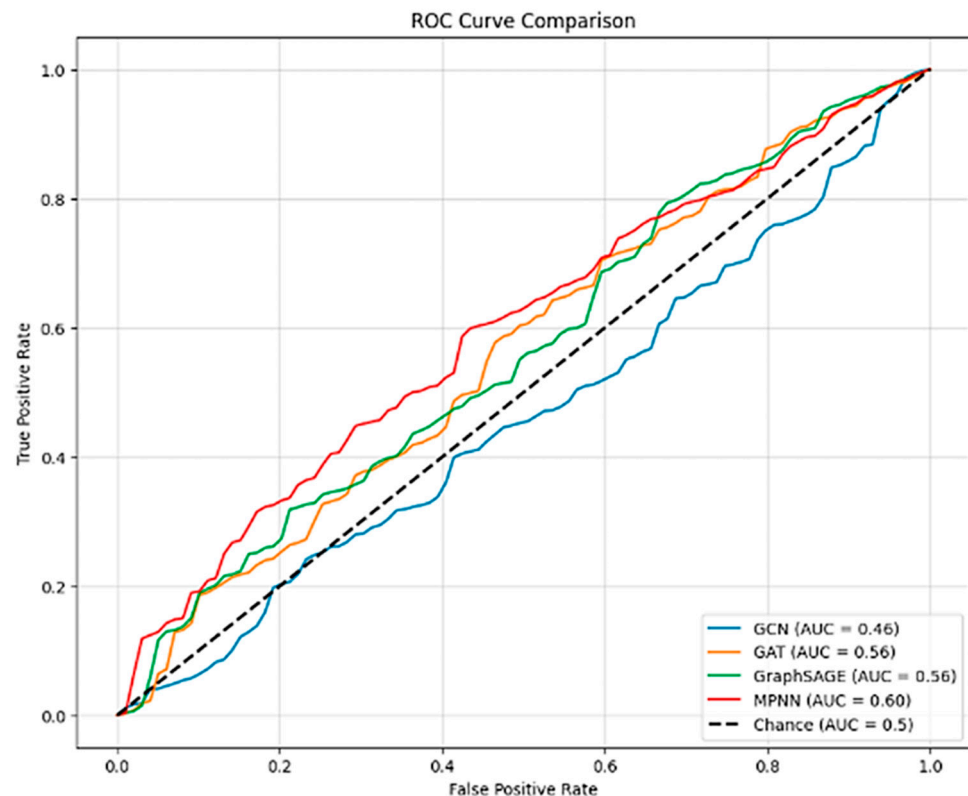


Figure 6. ROC curve comparison.

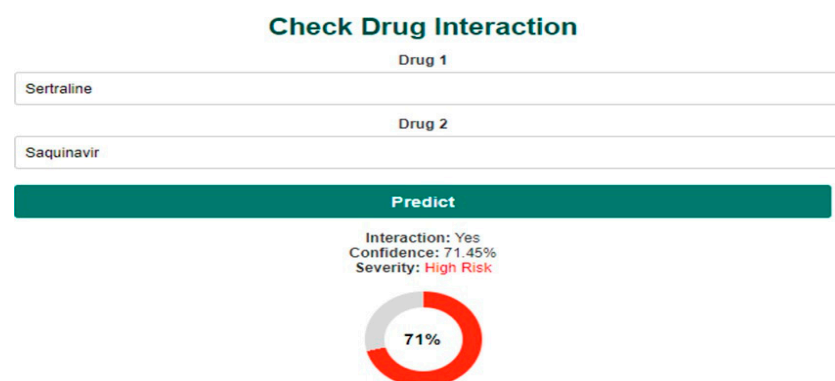


Figure 7. Interactive results.

Drug–drug interaction (DDI) prediction has seen significant development in recent years with machine learning (ML), as well as graph neural network (GNN) integration. These approaches are designed to mitigate the shortcomings of conventional rule-based and feature-driven methods that involve using graph-based ways for a more subtle representation of drugs and their interactions. This study builds on existing research in this field by adopting GraphSAGE as its main framework for ensuring scalability, interpretability, real-world usability, and enhancing the overall robustness of the predictions produced.

The prediction of drug–drug interactions (DDIs) through computational models has become crucial. The danger of having adverse DDIs in healthcare systems increases rapidly as polypharmacy, which is the practice where multiple medicines are prescribed to manage comorbidities, is being adopted. This requires sophisticated techniques that surpass traditional pharmacovigilance methods. This research aims at addressing this issue by performing predictive modeling with GNNs and creating a user-friendly system for healthcare providers.

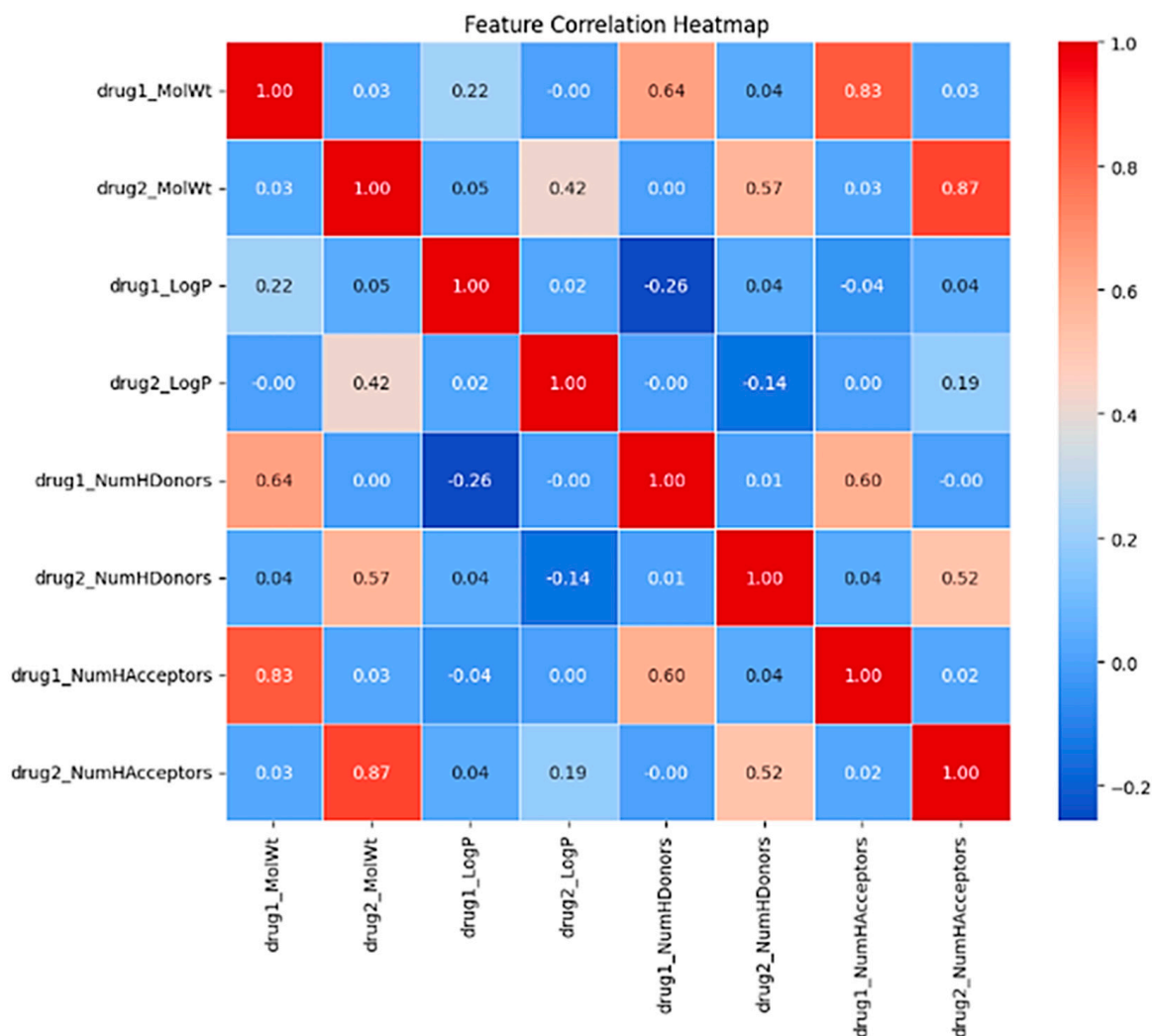


Figure 8. Feature correlation heatmap.

The importance of this study lies in its diverse nature. By utilizing contemporary GNN architectures together with strong molecular feature extraction techniques, it offers a scalable and understandable DDI prediction solution. RDKit and PubChemPy among other tools integrated into the system's pre-processing pipeline guarantees data quality and model reliability. Moreover, presenting it as a web application demonstrates how practical it is since healthcare providers can utilize real-time predictions.

This section synthesizes findings from the research, contextualizing them within the broader landscape of pharmacological research and clinical practice. It discusses what these findings mean, assesses limitations of the system, and presents future directions for inquiry into this subject matter. By relating findings to the existing literature, this section reiterates what was achieved and areas where more studies need to be performed.

The study highlights the transformative potential of GNNs in healthcare, serving as a guide for adding AI-driven systems into clinical workflows. More than that, this validates the feasibility of the proposed system and opens doors for further development in predictive pharmacology and personalized medicine.

5. Conclusions and Future Work

The comparison of GNN architectures has shown their individual strengths and weaknesses for future use. GraphSAGE has shown remarkable scalability that enables the model to process large and complex datasets effectively. Its ability to sample and aggregate neigh-

borhood features was instrumental in capturing indirect drug interactions. Conversely, the GAT had good interpretability, as it used attention mechanisms which specify the importance of some interactions. On one hand, the GCN was effective for direct interactions but failed with more complicated relational data structures. The MPNN was adept at catching intricate dependencies but faced problems with computational efficiency and overfitting. These findings underscore how important it is to choose a model architecture depending on specific requirements such as scalability, interpretability, or computation limitations of an application involved in drug discovery and interaction prediction.

This research makes a significant contribution to the field of AI-driven healthcare. This investigation shows that it is possible to use GNNs for predictive pharmacology by overcoming limitations in traditional mechanisms such as rule-based systems and basic machine learning models. The knowledge gained from this work will serve as a guide for future studies, underlining the importance of molecular data integration, advanced preprocessing, and model selection. Moreover, through success achieved by the program, the possibility to address other complex issues in pharmacology like drug discovery or individualized medicine has been confirmed.

Integrating multidrug forecasts into an existing platform would involve rigorous testing and validation. Such partnerships with pharmaceutical firms and regulatory agencies will speed up this process, ensuring that the system is in line with clinical standards. This aspect has the potential to disrupt the prescription of multi-drug regimens through making personalized medicine more accessible.

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